
Generative Latent Priors for Interpreting and Steering Protein Language Models

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Abstract

Sparse dictionaries are the dominant tool for interpreting Protein Language Models (PLMs). Generative Latent Priors (GLP) are a complementary approach: a diffusion model that learns the distribution of a frozen model’s activations directly, imposing no assumptions. We port GLP to ESM-2 activations (UniRef50) and evaluate it at two host scales (ESM-2-8M and -650M), both as a lens for PLMs representations and as a prior for editing them. Generation quality scales smoothly with depth and compute. As a *lens*, single GLP “meta-neurons” are the strongest probes of discrete Swiss-Prot concepts at both hosts, beating InterPLM’s released SAE features, and lead on continuous properties at 8M though not at 650M; falsification tests show this is mostly concentration, not new information, though a handful of concepts resist every sparse dictionary combination. As an *editing prior*, the on-manifold correction makes diff-of-means steering of three biochemical properties effective and restores on-manifoldness, but only in isolation: re-embedding the sequences actually produced, the correction no longer helps and buys no structural plausibility. This is mainly reproduced in additional SAE-based clamping structural steering experiments, where only Helix motif succeeds. The naturalness failure is traced to the discrete decoder rather than the prior, highlighting what the method must clear before it can serve for both *discovery-by-editing* and *discovery-by-exploration*.

1 Introduction

While protein language models (PLMs) have transformed sequence modeling in biology, interpreting the internal representations remains a fundamental challenge. In the activation space, the dominating methods rely on sparse dictionary learning, training sparse autoencoders (SAEs) on ESM-2 [7] activations and recovering features aligned with Swiss-Prot annotations [12, 4, 1]. The Generative Latent Prior (GLP) [8] is a complementary *generative* approach recently introduced for LLMs: rather than decomposing activations into a sparse dictionary, it trains a diffusion model on a frozen host’s activations, learning their distribution directly with no linearity or sparsity assumption.

We port GLPs to PLMs and evaluate them as a lens and an editing prior. Our contributions are: (i) trained flow-matching GLPs whose generation quality scales smoothly with depth and compute at both host sizes; (ii) a single-unit probing analysis in which GLP meta-neurons mostly isolate Swiss-Prot concepts better than InterPLM [12] features at both hosts, with additional falsification tests showing this is mostly concentration; and (iii) a two-mechanism, three-property test of GLP as an on-manifold editing prior, where the correction makes steering effective but not natural once decoded, a failure localised to the discrete decoder rather than the prior. Together they depict GLP’s role and limitations for control and discovery in PLMs; Fig. 1 overviews the pipeline.

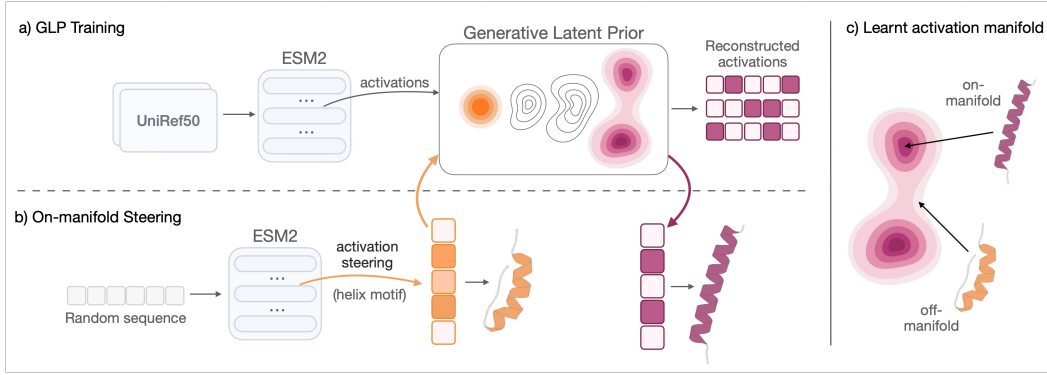


Figure 1: Pipeline overview. (a) GLP trains by flow matching on ESM-2 activations from UniRef50 to learn their manifold. (b) Steering pushes an activation off that manifold toward a target property. (c) GLP correction re-noises and denoises the edit back onto the manifold.

2 Method

2.1 Generative Latent Priors on ESM-2 activations

GLP substrate. ESM-2-8M / ESM-2-650M ($d=320$ / 1280) are frozen throughout. We pass UniRef50 sequences and read the residual stream after layers 4 and 18 respectively, both being layers InterPLM released SAEs for, so both methods are comparable (Appendix B.1). We train GLPs on 100M (8M/650M) and 500M (8M) activations from 5M random proteins.

Architecture. Let $h \in \mathbb{R}^d$ be one standardised per-residue activation from that stream, drawn from an unknown distribution $p(h)$. A GLP is a flow-matching model of $p(h)$: it learns to map Gaussian noise into a realistic on-manifold activation. Training corrupts a real h toward noise ϵ by a random amount t , giving $h_t = (1-t)h + t\epsilon$, and teaches a network u_θ to predict the direction that undoes the corruption,

$$\mathcal{L}(\theta) = \mathbb{E}_{h, \epsilon, t} \|u_\theta(h_t, t) - (\epsilon - h)\|^2, \quad (1)$$

so following $-u_\theta$ from pure noise ($t=1$) generates an activation, and following it from a partly-corrupted edit ($t_0 < 1$) repairs one. Here u_θ is a stack of SwiGLU MLP blocks, depth 3–24; its per-block gated activations, read at a fixed $t=u$, are the *meta-neurons* we probe. Full training hyperparameters are in Appendix B.2. Generation quality is Fréchet Distance (FD) to held-out real activations against a real-vs-real lower bound (Appendices B.2, A.2).

2.2 Concept probing: what do meta-neurons isolate?

We study whether GLP’s meta-neurons acting as *single* units isolate biological and biochemical concepts. We test 104/166¹ discrete residue-level Swiss-Prot annotations and 10 continuous whole-sequence properties across four unit representations: GLP meta-neurons, InterPLM’s SAE features, ESM’s own feedforward neurons, and the raw residual stream. Probes are trained on half the Swiss-Prot evaluation proteins (inner 5-fold CV selects the unit and ℓ_2 strength, giving the *validation* AUC) and scored on the disjoint held-out half, which is also excluded by accession from GLP’s training activations (Appendix B.1).

For **discrete concepts** we probe at residue level in three steps: (1) on the train split, shortlist units whose activations best separate concept-containing (positive) from concept-lacking (negative) residues; (2) fit each as a 1-D logistic probe and keep the highest *validation* AUC; (3) report that unit’s AUC on a held-out *test* split. Negatives come from the *same* proteins as positives, so a unit can only win by recognising the concept, not protein identity.

¹Frozen before probing: concepts must pass InterPLM’s frequency filter (>25 domain instances or $>1,500$ AAs) and a min-positives threshold on the test split, then span easy-to-hard SAE tractability per category, scored by the SAE. Yields 104 concepts at 8M, 166 at 650M (Section 3.2).

As falsification tests, we check whether each winning meta-neuron is linearly recoverable from the full 10,240-feature SAE dictionary, and whether a joint probe over the SAE’s top- k features ($k \leq 20$) matches GLP on the concept (Appendix D).

For **continuous properties**, we probe ten protein-level properties on whole-sequences with the same per-unit selection procedure: GRAVY (hydrophobicity), molecular weight, isoelectric point, aromaticity, instability index, charge at pH 7, aliphatic index, helix fraction, turn fraction and sheet fraction. Each label is computed directly from the amino acid sequence. In this case, we report test Spearman ρ instead of AUC, since a continuous property has no positive/negative split to define.

2.3 On-manifold steering

Editing an activation to bias generation can push it off-manifold, degrading the sequences subsequently decoded from it. A GLP that has learned the distribution of real activations offers a way to address this: project the edit back onto the manifold before decoding. This raises two questions we measure separately: does the correction restore *on-manifoldness*, by the GLP’s own diffusion loss? And does an on-manifold activation decode into a structurally plausible protein, by ESMFold pTM [7] We evaluate both under two edits sharing a mechanism (Algorithm 1): a masked-Gibbs resampling with the correction applied at every step or not at all (“naive”). All steering experiments reported use ESM-2-8M.

Difference-of-means property steering. Following Huang et al. [5], the steering direction v is the mean-activation difference between the top and bottom UniRef50 quartiles of a whole-sequence property. We add it to the hooked activation and renormalise, $\tilde{h} = (h + \alpha v) \|h\| / \|h + \alpha v\|$, so $\alpha=0$ is an exact identity, then regenerate the sequence over 30 rounds that each resample 10% of positions from the steered activation. We steer three properties (hydrophobicity, charge, instability [2]) in two directions: “up” from held-out low-scoring proteins and “down” from high-scoring ones. Each property uses 40 starting proteins per direction and 7 steering strengths, with and without the correction, for 1,120 generations. We report the **divergence** (up minus down): a genuine direction pushes the two apart, while an artifact shifts both alike and cancels. Each direction is first differenced against its own $\alpha=0$ generations, because the correction moves the property score even at zero steering (Appendix C.2).

Single-unit structural steering instead overwrites one interpretable SAE latent, $z_k \leftarrow \alpha$, and reconstructs the activation from the distorted dictionary. Following [13] we start from short random sequences, clamp and resample the whole sequence. We do 100 independent rounds, keeping the round maximising z_k . A continuous property has no single unit to clamp, but a structural concept the SAE isolates as roughly one feature can be pushed directly: for Helix, Beta strand, Coiled coil and Zinc finger we clamp the feature with the highest feature–concept F1 against Swiss-Prot labels. Success is a concept hit in the finished sequence, so we fold with ESMFold and detect Helix and Beta strand with DSSP, Coiled coil with pepcoil, and Zinc finger with a PROSITE [11] scan corroborated by a Pfam zf-C2H2 HMM alignment [10]. Since that call never fires, we also score every sequence continuously against the same profile (Appendix C.8).

3 Results

3.1 Generation quality scales with compute

GLP trains stably, converging well below the trivial-predictor floor, and its samples fill the real activation distribution’s principal subspace as sampler steps grow (Appendix A). Two properties emerge across a grid of four depths and two data scales. First, the diffusion loss follows a compute trend: plotted against $C=6ND$ [6], all eight runs collapse onto one curve (Fig. 2a), so the loss is set by the total compute budget, not by how it is split between depth (model size) and data. Second, Fréchet Distance to real activations improves with depth and this depth-scaling reproduces on ESM-2-650M. Because absolute FD is not comparable across hosts of different width, we read it on the FD lower-bound ratio, which normalises each host by its own real-versus-real floor (Fig. 2b).

Table 1: 1-D probing, select-on-valid/report-on-test. Discrete: mean test AUC, 95% bootstrap CI. Continuous: mean test Spearman ρ over 10 properties.

Representation	Discrete AUC		Continuous ρ	
	8M ($n=104$)	650M ($n=166$)	8M ($n=10$)	650M ($n=10$)
GLP	0.897 [0.879, 0.914]	0.938 [0.925, 0.950]	0.663 [0.611, 0.720]	0.545 [0.485, 0.605]
SAE [12]	0.842 [0.821, 0.862]	0.890 [0.872, 0.907]	0.605 [0.562, 0.653]	0.569 [0.497, 0.645]
ESM FFN	0.876 [0.858, 0.894]	0.841 [0.824, 0.858]	0.563 [0.524, 0.603]	0.377 [0.317, 0.429]
Res. Stream	0.827 [0.808, 0.846]	0.851 [0.833, 0.868]	0.469 [0.407, 0.532]	0.362 [0.300, 0.416]

3.2 Concept information concentrates in single meta-neurons

Meta-neurons are the best unit probes on most concepts. Table 1 reports mean test AUC over each host’s own qualifying Swiss-Prot concepts and mean test Spearman ρ over 10 continuous properties; GLP meta-neurons lead every baseline in every column. We test each GLP-vs-baseline margin as a matched pair and a paired bootstrap over these differences (10,000 resamples) gives the 95% CIs. At 8M, GLP wins on the majority of individual items against each baseline (94/104 concepts vs. the SAE, 83/104 vs. ESM’s own feedforward neurons, 98/104 vs. the raw residual stream, and 8/10, 8/10, 10/10 respectively on the continuous properties). GLP also leads four of five concept categories at 8M (per-concept and per-category numbers in Appendix D).

The discrete lead holds at scale. Repeating the single-unit comparison on ESM-2-650M (24-block GLP, layer 18) reproduces the discrete result (Table 1): a GLP meta-neuron again isolates concepts better than every baseline, winning 152/166 concepts against the SAE, 160/166 against ESM’s feedforward neurons, and 158/166 against the raw residual stream (all $p < 10^{-25}$). The two hosts use different concept pools (166 vs 104), so absolute AUCs are not comparable across hosts, but the within-host GLP-over-SAE margin is: +0.048 AUC at 650M, matching +0.055 at 8M. The lead is specific to discrete concepts: on the ten continuous properties GLP leads at 8M (0.663 vs 0.605) but not at 650M, where the SAE nominally edges it (0.569 vs 0.545, overlapping CIs).

Meta-neurons concentrate what the sparse dictionary spreads. Winning on a single unit could just mean GLP found a better rotation of a subspace the SAE already spans, and at 8M two tests say it largely does. Regressing each winning meta-neuron on the full SAE dictionary recovers it moderately (median $R^2 = 0.76$; 11 of 94 below 0.6), and a joint probe over up to 20 highest-AUC SAE features closes the gap for 79 of 94 concepts, so single meta-neurons hold the capacity of several dictionary atoms. The two tests disagree on *which* concepts are the exception: of the 11 units with $R^2 < 0.6$, only one is among the 15 the joint probe never reaches, so R^2 predicts neither membership nor margin (Appendix D). Of those 15, 13 fall short by noise-level margins (median +0.008 AUC), leaving Active site_Proton acceptor (+0.14) and Domain_Ubiquitin-like (+0.09) as candidates for concept-level structure the dictionary does not jointly span.

3.3 Correction succeeds but naturalness is compromised by decoding

On-manifold correction enables effective property steering. The naive arm (no correction) barely separates the up and down directions except at the strongest perturbation, whereas the correction separates them cleanly on all three properties across most of the grid (Table 2; full dose-response in Appendix C.2). Nevertheless, the correction carries a fixed compositional bias (Leu/Val enrichment, present even at zero steering, with no shift in charge), so we report every effect of it (Appendix C.3).

Structural steering is concept-dependent and locality-limited. Clamping a single SAE feature toward four structural concepts does not reproduce the latter result, as only Helix improves consistently. The apparent Coiled coil gain (6 $\rightarrow$ 33%) is simply helix formation scored as hits, Beta strand falls under correction (30 $\rightarrow$ 14%) and Zinc finger yields no motif (though a continuous HMM score shows a shift toward zf-C2H2 character). This is what a per-residue intervention can reach: while a helix is set by local propensities

Table 2: Steering divergence (up – down) at $\alpha=0.257$ (units of $\|h\|$); “SD” rescales by each property’s UniRef50 spread. Mann–Whitney, $n=40$ /direction; full sweep in Table 4.

property	natural SD	naive		GLP-corrected	
		divergence	p	divergence	SD units
hydrophobicity (GRAVY)	0.456	+0.08	0.44	+1.58	+3.46
charge / residue	0.060	+0.06	0.02	+0.22	+3.61
instability index	15.73	+5.30	0.19	+25.0	+1.59

within a short window, the rest need residues far apart in sequence to pair, harder to generate. Feature selection is also indirect: we select the clamp target by detection F1, assuming the best identifier is the best controller, yet the strongest detector (F1-0.907 for Zinc Finger) never finds a hit, so better, more sophisticated selection methods might reach concepts this rule cannot. Fold confidence follows the pattern, rising with the correction only where steering succeeds. All results are reported in Appendix C.8.

Decoding, not the prior, limits foldability and on-manifold restoration. Previous steering studies reports concept efficacy but not folding ([13, 5]); we fold every generation. The decoding loop drives every starting protein to a common, mediocre fold confidence regardless of where it started (Fig. 10, Appendix C.6), and both arms (naive vs corrected) land together, so the correction neither helps nor hurts foldability. The cause is the masked-Gibbs decoder, not GLP (Appendix C.4). On-manifold restoration is no better through decoding. On isolated steered activations, correction passes restore GLP’s diffusion loss to near the unsteered level (Appendix C.5). However, re-embedding the finished sequences, the corrected arm is not more on-manifold for both steering methods, plausibly because the corrected activation only picks amino acids and re-embedding returns whatever ESM computes, unconstrained by GLP. GLP’s typicality is thus not a proxy for sample quality.

4 Conclusions

In this study we port GLPs to PLMs, and evaluate them as a lens for reading PLM’s internal representations and as a prior for editing them. As a lens, a single GLP meta-neuron out-probes mostly every baseline (the SAE, ESM’s own feedforward units, the raw residual stream) at both ESM-2-8M and -650M, with some exceptions, but our falsification tests read most of this as *concentration*, not new information. GLP is thus a faithful, better-reading lens, and the concepts no dictionary combination matches are the promising leads for representation discovery. As an editing prior, the correction is mostly effective across biochemical properties and some restrained structural motifs, such as Helix. However, although the on-manifold edit has great effect in isolation, neither post-correction foldability nor on-manifoldness survive the decoding step in the implemented masked-Gibbs algorithm (Section 3.3). This is a structural limit on *discovery-by-editing* when using a continuous prior to steer generations.

Our study is bounded to one hook layer per host, correlational probing on a domain/motif-heavy pool and steering restricted to one layer and the masked positions. Future work splits along the same two roles. On the lens side, meta-neurons are read only by 1-D probing; automated interpretation of individual units and of the manifold’s local geometry could turn the leads above into defined concepts. On the editing side, foldability may improve with a better decoding algorithm (beam search already folds better per edit than masked-Gibbs, Appendix C.7), but the lost manifold restoration will not, since it comes from discreteness itself: once the head samples amino acids, the correction that produced them is gone. Fixing that needs a host whose output stays continuous throughout (structural or latent diffusion), or an edit constrained inside the sampler itself (either clamping a meta-neuron to a target value or gradient-guiding the denoising trajectory toward one) so the result is on-manifold by construction. Here, the correction pulled toward high density for control and discovery-by-editing; the same density estimate could instead detect and hold activations at the boundary of typicality, turning it into a tool for actual novel discovery.

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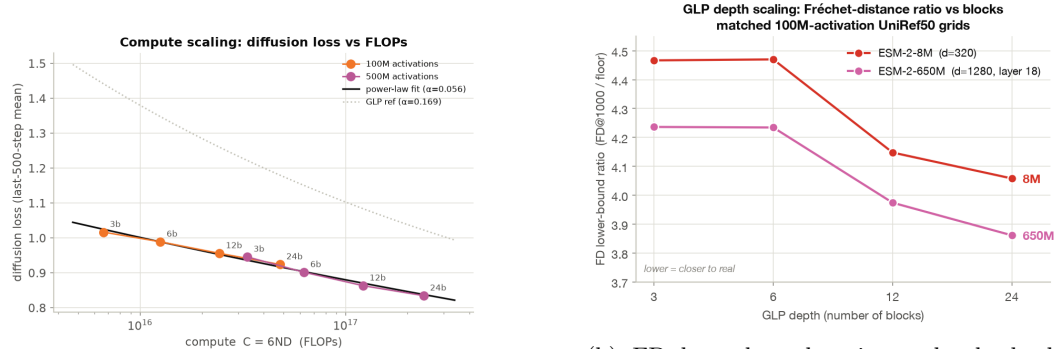
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A Scaling and evaluation metrics

A.1 Reading the compute-scaling figure (Fig. 2a)

Training a model costs *compute*, the total amount of arithmetic performed. A standard estimate [6] is $C = 6ND$, where N is the number of model parameters (how large the model is) and D is the number of training tokens it processes (here, activation vectors seen, i.e. our data size). The factor 6 is a fixed bookkeeping constant for one forward-plus-backward pass per token. So a model accumulates more compute if it is *bigger* (larger N) or trained on *more data* (larger D). Fig. 2a plots this compute C on the horizontal axis (log scale: each gridline is $10\times$ more) against the final training loss on the vertical axis (lower is better). Each point is one of our eight trained models at ESM2-8M scale. The key observation is that they fall on a single downward line: the loss depends essentially on the *total* compute, not on *how* it was spent, a small model on lots of data reaches about the same loss as a bigger model on less data, provided the compute matches. Such a relationship is called a *scaling law*.

We summarize the line by fitting $L(C) = E + AC^{-\alpha}$. Two numbers carry the meaning. α (“alpha”) is the slope: how quickly the loss falls as compute grows, a larger α means faster improvement per unit of compute. E is the *irreducible* loss, the floor you would approach with infinite compute (A is just a scale constant). We find $\alpha \approx 0.056$. The original GLP work, on language-model activations, reported $\alpha \approx 0.169$: their loss fell faster with compute, but our entire curve sits lower, meaning our protein-activation models reach a lower loss at the same compute. Because our compute range is short (~ 1 decade), the data does not reach far enough toward the floor to estimate E reliably, so we report only the slope and treat the fitted curve as a descriptive guide rather than a precise law.



(a) Compute scaling of the diffusion loss.

(b) FD lower-bound ratio vs depth, both hosts.

Figure 2: Generation quality scales with compute. (a) Smoothed final loss of all eight (depth $\times$ data) runs collapses onto one power law in $C=6ND$; our protein-activation curve lies below the published LLM-activation reference (dotted). (b) FD lower-bound ratio (FD@1000 / real-vs-real floor; lower is better) vs GLP depth, on matched 100M-activation grids: the shape ports across a $160\times$ host change (flat below six blocks, then improving) and ESM-2-650M is uniformly lower. Training curves, FD grid, and scaling table: Appendix A.

A.2 Fréchet Distance and its sampler dependence

FD compares two point clouds (here, generated vs. real activations) by their mean and covariance, lower is better, and it is 0 only when the two match in those first two moments. Because our generator is a diffusion sampler, its output quality depends on the number of integration steps: too few steps and FD mostly reports *sampler* error, not model quality. Fig. 4 makes this explicit, FD falls steeply from 20 to 1000 steps, so every FD number must be stated with its step count (we use 1000). We also report FD as a *ratio* to the FD(train, val) real-vs-real lower bound, which is the smallest FD achievable from finite samples of the

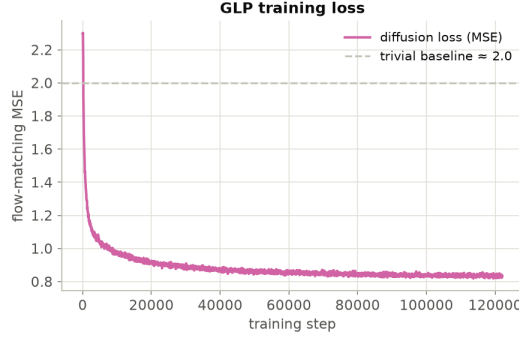


Figure 3: Training loss (24-block, 500M). GLP converges below the trivial-predictor floor (≈ 2.0).

Table 3: Depth $\times$ data-scale grid, generation quality only. N = parameters, $C = 6ND$, FD at 1000 sampler steps; ratio is FD / real-vs-real lower bound; loss is the mean of the last 500 training steps (the per-step flow-matching loss is noisy; Appendix A.1). The ESM-2-8M and ESM-2-650M hosts occupy different-dimensional activation spaces ($d=320$ vs. 1280), so absolute FD and loss are not comparable across the two host blocks; only the FD ratio is. The 650M grid is at the 100M data scale only.

Data	Blocks	Params	$C = 6ND$	FD@1000	FD ratio	Loss
<i>ESM-2-8M host ($d=320$)</i>						
100M	3	11.1M	6.7×10^{15}	4.27	$4.47 \times$	1.017
100M	6	20.9M	1.3×10^{16}	4.28	$4.47 \times$	0.989
100M	12	40.6M	2.4×10^{16}	3.97	$4.15 \times$	0.956
100M	24	80.0M	4.8×10^{16}	3.88	$4.06 \times$	0.924
500M	3	11.1M	3.3×10^{16}	3.54	$3.33 \times$	0.946
500M	6	20.9M	6.3×10^{16}	3.52	$3.31 \times$	0.902
500M	12	40.6M	1.2×10^{17}	3.23	$3.04 \times$	0.863
500M	24	80.0M	2.4×10^{17}	2.58	$2.42 \times$	0.835
<i>ESM-2-650M host ($d=1280$, layer 18); absolute FD/loss in this host's own space</i>						
100M	3	177M	1.1×10^{17}	1346.8	$4.24 \times$	1.038
100M	6	334M	2.0×10^{17}	1346.1	$4.23 \times$	0.974
100M	12	649M	3.9×10^{17}	1263.4	$3.97 \times$	0.919
100M	24	1.28B	7.7×10^{17}	1227.4	$3.86 \times$	0.877

true distribution; a ratio of 1 would be perfect, and the ratio is comparable across models trained on the same data.

B Data pipeline and reproducibility

B.1 Hook parity and the activation store

GLP is only a faithful model of “what ESM-2 computes” if it trains on exactly the tensor of interest. We read `hidden_states` [4] (the residual stream after transformer block 4) with `cls/eos` tokens dropped, which is byte-identical to what InterPLM’s released SAE consumed; a unit test asserts equality against InterPLM’s own embedder. Activations are extracted once per data scale into a memory-mapped `float16` store with a fitted standardizer (subtract mean, divide by standard deviation per feature), and all model/depth variants at that scale read the same store. Standardization is applied per batch at train time, never materializing the whole store in `float32`.

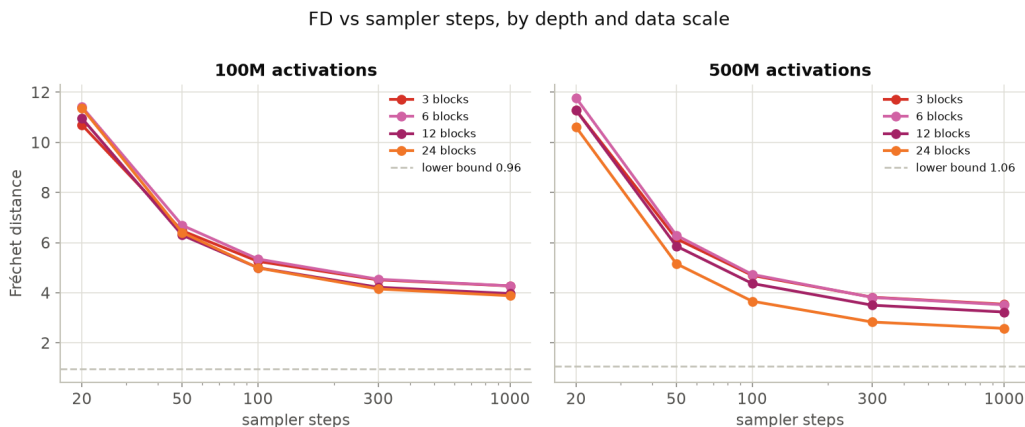


Figure 4: Fréchet Distance vs. sampler steps, by GLP depth and data scale (shared y -axis). FD falls with sampler steps everywhere (quote FD with its step count). Depth lowers FD monotonically, and the gain is larger at 500M than at 100M, more data lets extra depth pay off. The shallowest cells (3 vs. 6 blocks) are near-identical at both scales; separation appears at 12–24 blocks. Dashed lines: real-vs-real lower bound.

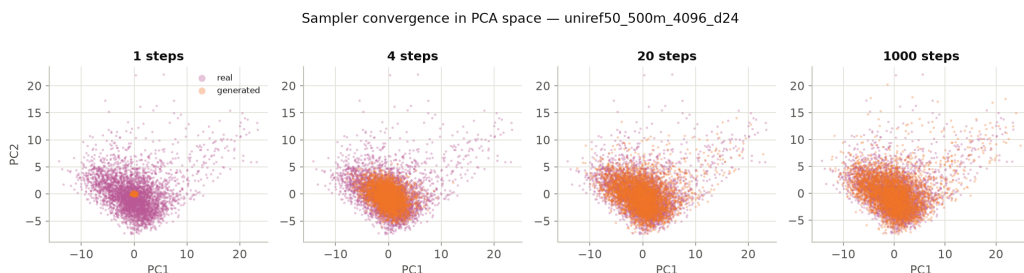


Figure 5: Sampler convergence in PCA space (24-block, 500M): generated activations (orange) refine from a collapsed blob at 1 step to matching the real distribution (purple) at 1000 steps, on the real data’s top-2 principal components.

Training data sampling. The training pool is a one-pass reservoir sample (Algorithm R, fixed seed) of 5M sequences from the local UniRef50 FASTA, matching InterPLM’s “random sequences from UniRef50” protocol rather than a first- N slice (which cluster-ID order would bias). Eval accessions (Probing splits, below) are filtered out during this pass. The sampled sequences are then embedded at the hooked layer in sampling order and their per-residue activations streamed into the memmap store until it reaches the target count (10^8 or 5×10^8), truncating mid-sequence at that row; the remaining sampled sequences are unused. Because the sampling order is itself random, the truncated store is an unbiased per-residue subsample of the 5M-sequence pool. The standardizer is then fit on a 500k-row random subsample of the store.

Train/eval disjointness. The Swiss-Prot evaluation proteins are held out of GLP’s training activations: every evaluation accession is passed to `extract_activations.py`’s `--exclude-from` and dropped from the UniRef50 sample by accession. Without this the default REST queries overlap the evaluation set by $\sim 98\%$, which would let GLP fit activations it is later probed on while the SAE, trained on generic UniRef50, gets no such advantage. How those evaluation proteins are then split into probe-training and test sets is in Appendix D.1.

B.2 Scaling-grid reproducibility

Because the activation store depends only on the data (host model, hook layer, sequences, count) and not on the GLP architecture or batch size, the depth grid at a given scale shares one store; configs declare it explicitly so no cell re-extracts. Each run records its full resolved configuration and a per-launch manifest (SLURM allocation, git commit, submit arguments), and training is checkpoint/resumable, so any cell can be reproduced or continued from its recorded state.

Training hyperparameters. Every GLP is trained by flow matching (Eq. 1) for a single pass over its activation store (100M activations at ESM-2-8M and ESM-2-650M, 500M at ESM-2-8M) in fp32, with AdamW (learning rate 5×10^{-5} , PyTorch-default β and weight decay) under a cosine schedule with 1% linear warmup (initial and final learning-rate factors 0.01 and 0.1) and gradient-norm clipping at 1.0. The batch is 4,096 activations; the flow-matching time t is drawn uniformly on $[0, 1]$ at a 1,000-step resolution. Only `n_blocks` (depth 3–24) and the derived widths ($d_{\text{model}}=2d$, $d_{\text{mlp}}=4d$) vary across the grid at a given scale. Generation uses 20 Euler steps; FD uses 1,000 (Appendix A.2).

C On-manifold steering

C.1 The steering and decoding procedure

Algorithm 1 states the full procedure. Every generation in this paper is one call to it, and the naive and GLP-corrected arms differ *only* in whether line 8 executes.

Three details are worth reading off the pseudocode because they are where our setup departs from the obvious one. First, the intervention is applied to the *masked* positions only, at a single layer, and is re-applied from scratch every round; it is not a persistent edit to the model. Second, the correction operates on the steered activation before it is decoded, so what the language-model head sees is always a GLP sample, never the raw sum $h + \alpha v$. Third, the renormalisation in line 6 makes $\alpha=0$ an exact identity, which is what licenses the $\alpha=0$ run as the within-arm control used throughout Appendix C.2.

The off-manifold diagnostic of Appendix C.5 is the same two operators without the loop: it applies line 6 and then SDEdit to real held-out activations and measures GLP’s diffusion loss, with no masking, no decoding and no iteration.

C.2 What we measure and why

At $\alpha=0$ the diff-of-means intervention is an exact identity, so the naive and GLP-corrected arms differ *only* by whether the correction ran. They do not agree there: the corrected arm gains +1.15 ΔGRAVY against the naive arm’s +0.35, which is 65–97% of the corrected arm’s raw effect at $\alpha \leq 0.257$. A raw corrected-vs-naive comparison therefore credits the correction’s own behaviour to steering.

We use two derived quantities throughout, computed *within* a direction (pooling the two would average a rising trajectory against a falling one and cancel the effect). The **steering-attributable** effect differences each sequence against the *same starting protein’s* $\alpha=0$ run in the same arm and direction, removing both the correction’s drift and per-protein variation; it is tested paired (Wilcoxon). The **divergence** is the up-minus-down difference of that quantity; since the two directions use different held-out sequences, it is tested unpaired (Mann–Whitney).

Degenerate sequences ($\geq 30\%$ one residue, ≤ 8 distinct amino acids, or a ≥ 12 -long homopolymer run) are excluded from effect estimates and reported separately as a rate. Because that gate is applied after generation and complexity is itself affected by the treatment, we report every headline number both gated and ungated; they agree to $\leq 1\%$.

Making three properties comparable. Two choices are needed before the three runs can appear in one table. First, **charge is scored per residue**. Net charge is a sum

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Algorithm 1 On-manifold steering by masked-Gibbs decoding. M : ESM-2-8M; $h^{(4)}$: its layer-4 residual stream (the tensor GLP is trained on); u_θ : the GLP velocity field; S : the standardizer fitted on the activation store. Canonical Exp. B settings: $\rho=0.10$, $R=30$, $T=1.0$, $t_0=0.7$, $K=20$. The “down” arm is the same call with $-\alpha$, started from held-out proteins at the opposite extreme of the property.

Require: starting sequence x , steering vector v , strength α , mask fraction ρ , rounds R , temperature T , SDEdit level t_0 , Euler steps K , flag CORRECT

- 1: $k \leftarrow \max(1, \text{round}(\rho |x|))$
- 2: **for** $r = 1 \dots R$ **do**
- 3: $P \leftarrow k$ positions drawn uniformly without replacement from $1 \dots |x|$
- 4: $x[P] \leftarrow \text{<mask>}$
- 5: $h \leftarrow h^{(4)}(M, x)$ ▷ forward pass on the masked sequence
- 6: $\tilde{h}_P \leftarrow (h_P + \alpha v) \cdot \frac{\|h_P\|}{\|h_P + \alpha v\|}$ ▷ diff-of-means edit, norm preserved
- 7: **if** CORRECT **then**
- 8: $\tilde{h}_P \leftarrow \text{SDEdit}(\tilde{h}_P; t_0, K)$ ▷ project back onto the GLP manifold
- 9: **end if**
- 10: $\ell \leftarrow M(x)$ with $h_P^{(4)}$ overridden by $\tilde{h}_P$ ▷ second forward pass
- 11: $x[P] \sim \text{softmax}(\ell_P/T)$ restricted to the 20 amino acids
- 12: **end for**
- 13: **return** x

- 14: **function** SDEdit($a; t_0, K$)
- 15: $z \leftarrow S(a); \quad \epsilon \sim \mathcal{N}(0, I)$
- 16: $z \leftarrow (1 - t_0)z + t_0 \epsilon$ ▷ re-noise to time t_0
- 17: **for** $t = t_0$ down to 0 in K equal Euler steps of size Δ **do**
- 18: $z \leftarrow z - \Delta u_\theta(z, t)$
- 19: **end for**
- 20: **return** $S^{-1}(z)$
- 21: **end function**

over residues, so ranking proteins by it ranks them substantially by length and the diff-of-means vector would encode length as much as charge; GRAVY is already a per-residue mean and the instability index is already length-normalised. With the per-residue score the two contrast sets differ in mean length by +0.13, -0.03 and -0.13 pool standard deviations for the three properties, and length correlates $|r| \leq 0.053$ with all three scores across the pool, so no length matching was needed.

Second, α is expressed relative to the activation norm. The edit $h + \alpha v$ rotates h by an angle set by $\alpha \|v\|/\|h\|$, and the three vectors have norms 3.70, 2.94 and 3.33 against a mean $\|h\|$ of ≈ 28.5 ; a shared grid in raw units would therefore be three different perturbations. We sweep $\alpha \in \{0, 0.032, 0.064, 0.128, 0.257, 0.385, 0.642\}$ in units of $\|h\|$. Effects are reported in units of each property’s standard deviation across the 5,790-sequence UniRef50 pool the contrasts were drawn from (0.456 GRAVY, 0.060 charge/residue, 15.73 index), not the spread of the held-out starts, which is bimodal for the steered property by construction and would shrink exactly the on-target entries.

Table 4: Steering-attributable divergence (up – down) at every grid point, both arms. α is in units of $\|h\|$; SD rescales the corrected divergence by the property’s UniRef50 spread. Unpaired Mann–Whitney, $n=40$ per direction.

property	α	naive	p	GLP-corrected	p	SD
gravity	0.032	-0.171	0.17	+0.248	0.02	+0.54
gravity	0.064	+0.006	0.93	+0.551	1×10^{-5}	+1.21
gravity	0.128	+0.112	0.23	+0.969	7×10^{-10}	+2.12
gravity	0.257	+0.077	0.44	+1.581	2×10^{-13}	+3.46
gravity	0.385	+0.546	8×10^{-5}	+2.507	4×10^{-14}	+5.49
gravity	0.642	+1.400	1×10^{-10}	+3.382	2×10^{-13}	+7.41
charge	0.032	-0.006	0.90	+0.011	0.62	+0.18
charge	0.064	-0.007	0.62	+0.040	4×10^{-3}	+0.67
charge	0.128	-0.001	0.96	+0.092	5×10^{-9}	+1.54
charge	0.257	+0.055	0.02	+0.216	2×10^{-14}	+3.60
charge	0.385	+0.157	6×10^{-10}	+0.313	2×10^{-14}	+5.23
charge	0.642	+0.430	2×10^{-12}	+0.560	3×10^{-13}	+9.34
instability	0.032	-6.17	0.12	+6.77	0.02	+0.43
instability	0.064	+4.43	0.25	+11.73	2×10^{-3}	+0.75
instability	0.128	+3.68	0.32	+13.55	3×10^{-5}	+0.86
instability	0.257	+5.30	0.19	+25.04	7×10^{-9}	+1.59
instability	0.385	+15.65	2×10^{-4}	+39.48	1×10^{-11}	+2.51
instability	0.642	+17.01	5×10^{-5}	+64.93	4×10^{-14}	+4.13

C.3 The correction’s own concept drift

Fig. 7 shows the $\alpha=0$ arms by direction. Resampling alone reverts toward typical hydrophobicity: the naive arm moves +0.35 from hydrophilic starts and -0.28 from hydrophobic ones, flipping sign with the starting point, while the corrected arm does not (+1.15 and +0.54), a fixed bias rather than reversion toward the training distribution’s typical composition, a distinction a single-direction design cannot make. The bias is compositional: since GRAVY is a mean over Kyte–Doolittle values, it decomposes exactly as $\Delta \text{GRAVY} = \sum_a \Delta f(a) \text{KD}(a)$, and Leu (7.1→18.0%) and Val (5.1→13.6%, roughly twice natural abundance) together account for 57% of it, the rest from depletion of charged and polar residues.

The same bias, seen through three properties. Table 5 and Fig. 6 separate fixed bias from mean reversion for all three targets: the naive arm shows clean, near-symmetric mean reversion everywhere, while the corrected arm adds a fixed offset equal from both starting sets to within 3% on hydrophobicity and 12% on instability, and absent on charge, exactly what a single Leu/Val enrichment predicts, since those two residues are the most

hydrophobic on the Kyte–Doolittle scale and stabilising in the Guruprasad table, and the accompanying loss of Lys+Arg (4.0pp) and Glu+Asp (4.4pp) cancels in net charge. The strongest evidence for this is that the shift is near-identical across all three runs despite sharing no data, contrast sets, starting proteins, or steering vectors, only the same correction operator: Leu +5.9/+6.3/+6.7pp and Val +5.4/+5.7/+4.0pp for hydrophobicity, charge and instability respectively, with Glu, Lys, Arg and Asp each down 1.3–2.8pp. It is a property of SDEdit under this GLP, not of any steering direction.

Table 5: Concept change with *no* steering applied ($\alpha=0$), by arm and starting set, in each property’s own units. A fixed bias adds the same amount from both sets; mean reversion flips sign with the set and cancels in the corrected–naive difference, which is the last column (there expressed in natural SD).

property	naive		GLP-corrected		fixed bias (SD)
	from low	from high	from low	from high	
hydrophobicity	+0.347	−0.282	+1.145	+0.536	+1.77
charge	+0.055	−0.039	+0.063	−0.071	−0.20
instability	+14.99	−15.48	+4.66	−27.13	−0.70

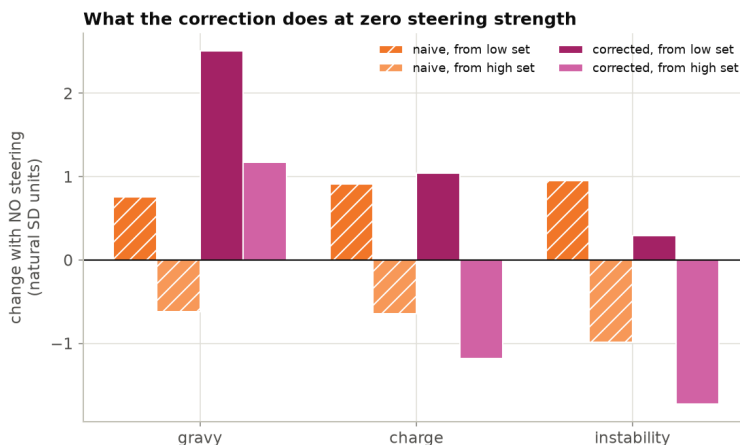


Figure 6: The $\alpha=0$ arms for all three properties, in units of each property’s natural spread. Hatched: naive. A bias that reverses with the starting set is mean reversion; one that does not is carried by the correction.

C.4 The naturalness metrics are unreliable, and differently so per property

Fig. 8 scores the sequences our degeneracy filter *rejects*, unambiguous non-proteins, against the sequences it keeps, for each steered property. pLDDT and pseudo-perplexity are gamed everywhere. pTM is not a reliable substitute: its verdict changes with the property (Table 6).

Table 6: ESMFold2 pTM on the sequences the degeneracy gate rejects versus those it keeps, by steered property (Mann–Whitney). “Gamed” means the metric scores known non-proteins better than our own clean output. pLDDT and pseudo-perplexity are gamed in all three ($p \leq 2 \times 10^{-2}$ and $p \leq 1 \times 10^{-12}$ respectively).

steered property	what is rejected	rejected	kept	p	verdict
hydrophobicity	Asp 21%, Thr 19%	0.275	0.268	0.98	indistinguishable
charge	Asp 24%, Glu 15%	0.180	0.246	4×10^{-4}	discriminates
instability	Val 43%	0.369	0.249	9×10^{-5}	gamed

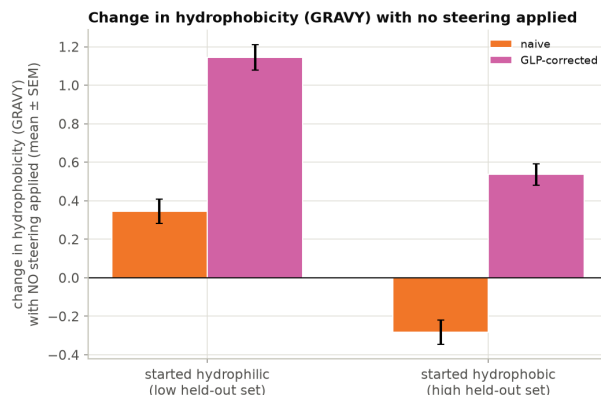


Figure 7: Change in hydrophobicity with *no* steering applied, by starting set. The naive arm reverts toward typical (sign flips with the starting set); the corrected arm pushes one way regardless.

The mechanism is visible in the second column. Charge’s degenerate output is poly-acidic, and poly-acidic chains are intrinsically disordered, so ESMFold correctly assigns them low confidence. Instability’s is 43% valine, which is what a confident β -sheet prediction looks like, so it scores above our own clean output. Whether a structural metric catches the junk therefore depends on whether *that* junk happens to be foldable, which is a property of the treatment, not of the metric, and no axis can be chosen safely in advance. A composition-based gate catches all three because it asks only whether the output is a sequence, not whether it is a good one. This is why the gate, and not any metric, is load-bearing in every number we report.

On a smaller run the rejected sequences scored pLDDT 81.0, above a natural ubiquitin control at 80. Two consequences. First, a quality threshold must be used as an *outcome* and not as a filter: selecting sequences with pLDDT ≥ 70 would retain 87% of the degenerate sequences and 12% of the clean ones, taking a set from 8% to 38% non-protein, and the two arms survive such a cut at different rates. We therefore report the fraction of *all* generations that are non-degenerate *and* above threshold. Second, on kept rows alone pseudo-perplexity correlates correctly with pLDDT ($\rho = -0.48$, $p = 4 \times 10^{-17}$), which reads as mutual validation but is shared bias, the correlation is computed after the filter has removed the rows that expose it. Metric behaviour should be checked on rejected rows, not only on kept ones.

Naturalness scores for sequences the composition filter kept vs. rejected

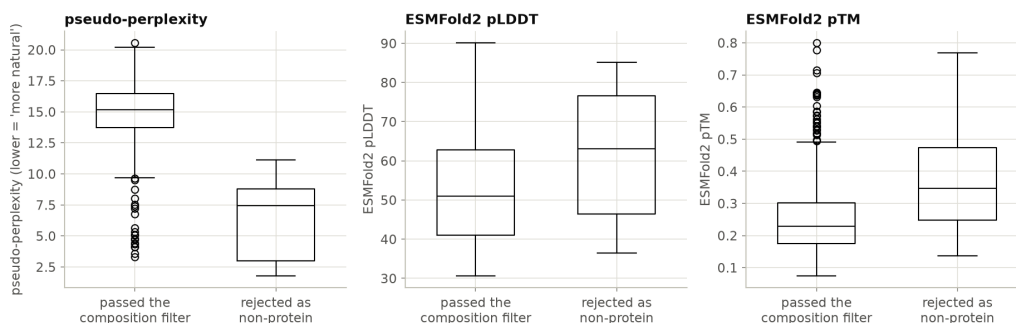


Figure 8: Naturalness scores for sequences the composition filter kept versus rejected, for the instability run, the case where pTM as well as pLDDT and pseudo-perplexity rates the rejected sequences above the kept ones.

C.5 The correction operator works; the decoder is what fails

Steering does move activations off the manifold, and monotonically in α : GLP’s diffusion loss on real held-out activations rises from 0.869 unsteered to 1.743 at the top of the grid (Fig. 9). Applying a single SDEdit pass to the steered activation with *no* generation returns it to 0.944, recovering 92%, 97% and 103% of the excess at the top three grid points. Values at or above 100% are real and mean the correction placed the activation in a *higher*-density region of GLP’s own distribution than the unsteered original; the same effect is why the corrected curve sits marginally below baseline at the bottom of the grid, where the excess to remove is ≈ 0 .

A wider sweep in raw units, reaching about $4\times$ the top of this grid, recovers 89% of an excess that by then has driven the loss to 3.18. The operator therefore handles heavily-steered activations; what degrades the output is decoding those activations back to amino acids and iterating that. Consistently, an apparent “breakdown” at the top of the grid in earlier runs was an artifact of the SDEdit noise level: moving t_{start} from 0.5 to 0.7 reduced degenerate output from 26% to 7.7% and made the dose-response monotone throughout.

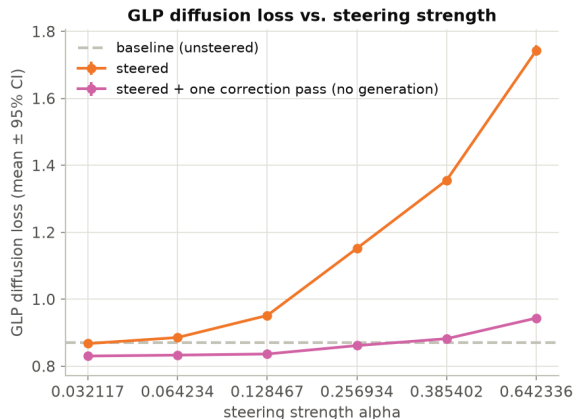


Figure 9: GLP diffusion loss against steering strength, before and after a single correction pass applied directly to the steered activation (no generation), for the hydrophobicity direction. α is in units of $\|h\|$, as everywhere else.

C.6 The decoding protocol, and what it costs

Fig. 11 shows the effect rising with α while structural confidence stays flat, for both arms, far below the starting proteins.

The decoding budget, not the mask/round split, governs the loss: at matched expected coverage (0.10), 0.02×5 , 0.05×2 and 0.10×1 give nearly identical pTM (0.364/0.364/0.366) and effect (+0.057/+0.081/+0.078). Fig. 12 plots both objectives against the fraction of residues resampled at least once: foldability stays within the starting proteins’ own interval up to $\sim 15\%$ coverage and declines beyond it, while the effect grows throughout, so the trade-off is between total rewriting and structure, not between mask fraction and round count.

The loop is a fixed point, not a constant penalty. With only the two hydrophobicity sets this looked like “masked-Gibbs roughly halves foldability”; six starting sets spanning pTM 0.369–0.650 show that was a coincidence of where those two sat. After the $\alpha=0$ naive pass all six land between 0.227 and 0.315 (Table 7, Fig. 10); regressing post on pre gives $\text{post} = 0.172 + 0.199 \text{ start}$ ($r=+0.80$), a fixed point at pTM 0.215 (-0.98 correlation between how well a set folds and how much it loses, better starting proteins simply have more to lose).

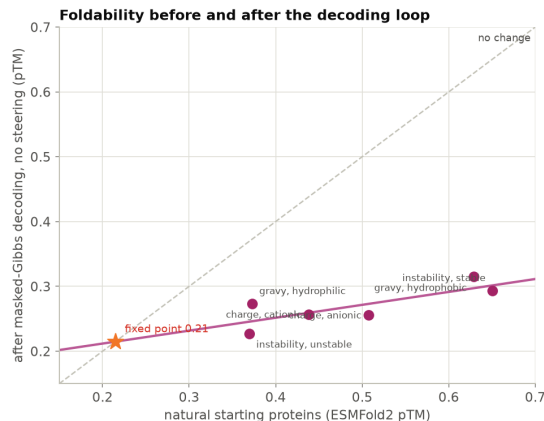


Figure 10: Foldability of the natural starting proteins against foldability after the decoding loop with no steering and no correction, for all six held-out sets. The dashed line is no change; the star marks where the fitted line crosses it. Both steering arms land at this same mediocre fixed point.

Table 7: Foldability before and after the decoding loop, with no steering and no correction ($\alpha=0$, naive arm), for all six held-out sets.

starting set	natural pTM	after decoding	change
instability, stable	0.650	0.294	−55%
hydrophobicity, hydrophobic	0.629	0.315	−50%
charge, anionic	0.507	0.256	−50%
charge, cationic	0.438	0.257	−41%
hydrophobicity, hydrophilic	0.373	0.274	−27%
instability, unstable	0.369	0.227	−38%

Neither steering strength nor the correction itself costs further foldability. Comparing the corrected arm at $\alpha=0$ against the top of the grid, paired by starting protein: -0.004 pTM ($p=0.70$) for hydrophobicity, -0.025 ($p=0.18$) for charge, -0.031 ($p=0.014$, Holm-adjusted 0.042) for instability, only instability is marginally significant, and no pass rate moves. A separate, powered test of the correction itself (an earlier version of this work saw a weak signal that it costs a little foldability) also does not survive: pooling 235 paired $\alpha=0$ comparisons across the three properties gives -0.012 pTM (95% CI $[-0.029, +0.005]$, $p=0.43$, no property individually below $p=0.14$). We therefore make neither foldability-cost claim, beyond the fixed point itself.

Table 8 summarizes the two decoding settings used elsewhere in this appendix: a thorough decode (30 rounds, $T=1.0$) and a 6 $\times$ -cheaper one (5 rounds, $T=0.5$) that trades divergence for speed at a similar degeneracy rate; both sit well below the natural starting proteins’ own foldability.

Table 8: Two operating points, both at $n=40$ starting proteins per direction. pTM is measured on the $\alpha=0$ naive arm, i.e. the decoding loop alone.

setting	divergence ($\alpha=5$)	pTM (low/high-GRVY starts)	degenerate
natural starting proteins	n/a	0.373 / 0.629	0%
30 rounds, $T=1.0$	+3.38	0.275 / 0.319	2.7%
5 rounds, $T=0.5$	+1.56	0.332 / 0.435	3.4%

Two cautions. Lowering temperature while keeping many rounds does *not* help: at $T=0.5$ with 30 rounds the decoder mode-collapses (8/15 non-proteins), and since collapsed se-

quences score high pTM this looks like the best grid setting to any analysis that drops degenerate rows instead of counting them; and a foldability comparison must use the held-out set a run actually generates from (0.373 or 0.629), not their pooled average.

We also tried [5]’s relatedness-ranked site selection (mutating residues least aligned with the steering vector, instead of a uniform random subset): at matched edit budget it preserved twice as much of the starting protein but cost $\sim 6\times$ of the effect ($+0.97 \rightarrow +0.17$) with no significant foldability gain, plausibly because our single-layer, edited-positions-only intervention is too weak to move the residues the host model most resists changing ([5] steer every layer and token). We retain uniform random masking.

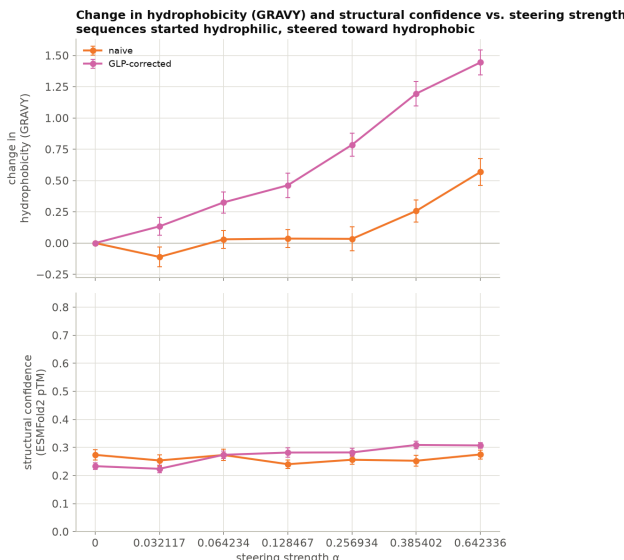


Figure 11: Effect and structural confidence against steering strength, sequences started hydrophilic. The lower panel is flat: the effect is bought without further structural cost, but from an already-degraded baseline.

C.7 Can a better decoder escape the fixed point?

The fixed point of Appendix C.6 raises whether the foldability loss is specific to uniform-random masked-Gibbs decoding or intrinsic to decoding steered activations at all. We tested two alternatives, both leaving the residues the host model is already confident about untouched, and both using the identical steering vector and correction operator; only the decoder changes.

Confidence-frozen remasking mode-collapses. Following the confidence-ordered unmasking of masked generative transformers [3], we remask each round only the positions the host model least endorses (lowest $p(x_i | \text{context})$) and freeze the rest. This collapses: 92.7% of outputs are degenerate (against 8.5% for uniform random masking), typically homopolymers, with a representative sequence 54–70% proline. Because ESM-2 is most confident about repetition, freezing its confident residues entrenches a growing repeat. The collapsed sequences score *higher* pTM (0.39 vs. 0.29) than the uniform-random decoder, but this is a folded poly-proline helix rather than a protein, and is caught only by counting degenerate outputs rather than filtering them.

Beam search over the steered pseudo-likelihood is the opposite failure mode.

We replace the resampling loop with the stochastic beam search of [9], adapted so the per-position marginals are the steered, GLP-corrected ones: from the real starting protein we make only E single-residue edits, each chosen to maximise the sequence’s steered pseudo-log-likelihood (every single-substitution neighbour is scored for free from one masked-marginal

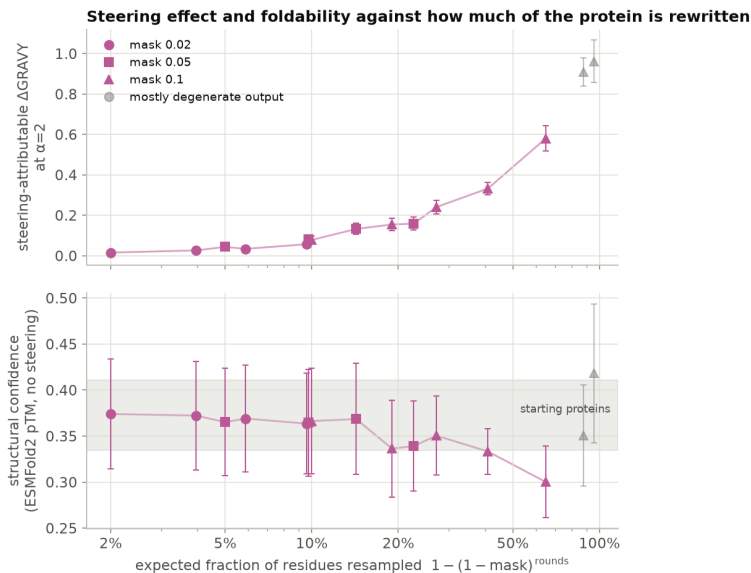


Figure 12: Steering effect (top) and foldability (bottom) against how much of the protein the decoder rewrites, $T=0.5$. Marker shape distinguishes mask fraction; the three series fall on a single curve, i.e. mask fraction and round count act only through their product. Shaded band: the starting proteins’ own pTM ($\pm$ SEM). Grey points are settings whose output is mostly degenerate and are excluded from the trend. This budget sweep predates the relative- α parameterisation and its axis label is in raw units; for the hydrophobicity vector its $\alpha=2$ is the same perturbation as the $\alpha=0.257$ used elsewhere in this paper.

pass). It never collapses: degeneracy stays at the natural-protein rate (2.5–4.6%) even under a $20\times$ steering push, and it keeps foldability near natural (pTM 0.44–0.49 at $E \leq 10$), but it steers only weakly.

Two levers, one curve. Raising α saturates almost immediately ($+0.11$ at $\alpha=1$ to $+0.17$ at $\alpha=20$, $E=10$); the edit budget E is the lever that works (Table 9), growing the effect while foldability declines gracefully and degeneracy stays low. The two decoders turn out to be the *same* efficacy/naturalness curve parameterised by edit count: masked-Gibbs performs ≈ 390 resampling events ($30 \text{ rounds} \times 10\% \times \sim 130 \text{ residues}$), and extrapolating the beam-search curve to that budget reproduces its corner: few edits are natural but weak, many are strong but degrade. At $\alpha=0$, where the arms differ only by decoder, the same starting proteins fold at pTM 0.48 through beam search and 0.29 through masked-Gibbs, the decoder, not the correction or the steering, sets the foldability. Neither decoder escapes the trade-off, though beam search is cleaner per edit (4.6% degenerate at $E=50$ vs. 8.5%) and may dominate it at matched effect, a direction we leave open.

Table 9: Beam search over the steered pseudo-likelihood, as a function of the edit budget E , against uniform-random masked-Gibbs. pTM is the $\alpha=0$ naive arm (the decoder alone); divergence is the attributable up-minus-down effect at the top of the grid; both at $n=40$ starting proteins per direction, hydrophobicity.

decoder	degenerate	pTM ($\alpha=0$)	divergence
beam, $E=5$	2.5%	0.486	+0.07
beam, $E=10$	2.5%	0.473	+0.08
beam, $E=25$	3.5%	0.438	+0.27
beam, $E=50$	4.6%	0.409	+0.51
masked-Gibbs, 30 rounds	8.5%	0.293	+3.99

C.8 The SAE-feature-clamp variant: four structural concepts

Instead of a diff-of-means direction, [13] clamp one SAE latent, $z_k \leftarrow \alpha$. We run this under the same decoder and naive/GLP-corrected split (Algorithm 1, the clamp replacing line 6) on Helix (feature 5355), Beta strand (2868), Coiled coil (813) and Zinc finger (1435, the C2H2 union), each clamped to the feature with the highest feature–concept F1. Each generation starts from a uniform-random 25–40-residue sequence and re-masks and resamples every position for 100 rounds, keeping the round that maximises z_k . $\alpha \in \{0, \dots, 5\}$ ($\{0, \dots, 8\}$ for Zinc finger), $n=100$ per cell. This run predates the decoding fixes of Appendix C.6 and should not be read against the main operating point.

The correction’s benefit is concept-dependent, unlike the uniform three-property result (Table 10). It raises the DSSP helix fraction ($0.54 \rightarrow 0.77$ at $\alpha=0$; the naive arm reaches ~ 0.73 by $\alpha=1$) and the `pepcoil` coiled-coil hit rate ($6 \rightarrow 33\%$ at $\alpha=5$), but the coiled-coil gain is `pepcoil` scoring long single helices as heptad repeats, not real pairing. Beta strand barely moves, and the correction pushes its strand fraction below naive for most α . Zinc finger produces no dual-verified C2H2 motif at any α (0 sequences here and across 18 further variants, $n=14,381$), yet on the graded PF00096 profile-HMM score the corrected arm exceeds naive at $\alpha \in \{1, 2, 4, 5, 6\}$ (Mann–Whitney, best $p=2 \times 10^{-5}$ at $\alpha=6$), driven by histidine enrichment (His per sequence $1.1 \rightarrow 4.3$ at $\alpha=4$) rather than motif completion. Fold confidence stays low for every concept (pLDDT in the 50s to low 70s), and the correction’s $\alpha=0$ pLDDT boost fades as α grows.

Why helix is the exception. The clamp sets one scalar per position, a compositional bias, and the decoder resamples each position from its own distribution, so it can change which amino acids appear but not how residues sit relative to one another. Local helix propensity survives that, so helix content rises; a β -sheet, a coiled coil and a zinc finger each need specific distant residues paired (two strands hydrogen-bonded, two helices in heptad register, two Cys and two His chelating an ion), which no per-position bias can arrange.

On-manifoldness does not reach the sequence. As with diff-of-means steering (Section 3.3), one correction pass returns the clamped activation close to GLP’s typical density in isolation, but that gain is lost at decoding: once the head has sampled amino acids, ESM re-embeds them and the activation is recomputed from the sequence rather than carried over, so the finished sequence is no more on-manifold than the naive arm’s. The limit is the discrete decoder, not the prior.

D Probing protocol, concept definitions, and per-concept tables

D.1 Discrete concepts: definition, selection, and probe

Concepts and splits. Each discrete concept is one residue-level UniProtKB feature annotation from Swiss-Prot, materialised per amino acid by InterPLM’s `extract_annotations` and named by its feature type and description (`Domain_SH3`, `Compositional_bias_Acidic_residues`, and so on); a residue is positive for a concept if it lies in an annotated span of it. The evaluation set is a 20,000-protein reviewed Swiss-Prot query filtered to proteins with an AlphaFoldDB model and at most 1,022 residues, split by protein into eight shards: 0–3 train the probes, 4–7 are the held-out test set, both disjoint from GLP’s training activations (Appendix B.1).

Selection (frozen before any GLP probing). We keep concepts that clear InterPLM’s frequency filter (>25 domain instances or $>1,500$ positive residues) and have ≥ 500 positive residues in the test shards, tag each into one of five biological buckets, and within each bucket keep concepts spanning the SAE’s single-feature tractability (its best per-feature validation AUC), so a GLP win cannot be a cherry-pick of SAE-hard concepts. The rule uses only the annotations and the SAE, never GLP, and yields 104 concepts at ESM-2-8M and 166 at 650M, dominated by the domain / motif bucket (a coverage limit noted in Section ??).

Probe. Per (concept, representation): shortlist the top $k=512$ units on shards 0–3 by $|\bar{f}_{\text{pos}} - \bar{f}_{\text{neg}}|$, fit each as a 1-D ℓ_2 -logistic probe with 5-fold cross-validation, keep the highest

Table 10: SAE-feature-clamp steering by α , naive/GLP-corrected, $n=100$ per cell. Structural column: DSSP secondary-structure fraction (per-chain residue fraction in that class) for Helix and Beta strand; EMBOSS `pepcoil` hit rate for Coiled coil; and, for Zinc finger, the fraction of sequences with any PF00096 zf-C2H2 profile-HMM alignment (a graded measure, no complete motif forms). pLDDT and pTM are naive/corrected. On the continuous Zinc-finger HMM score the corrected arm exceeds naive at $\alpha \in \{1, 2, 4, 5, 6\}$ (Mann-Whitney, best $p=2 \times 10^{-5}$).

α	structure (naive)	structure (corr.)	pLDDT (naive/corr.)	pTM (naive/corr.)
<i>Helix, feature 5355 (DSSP helix fraction)</i>				
0	0.54	0.77	49.7 / 59.8	0.153 / 0.251
1	0.73	0.78	59.7 / 62.3	0.270 / 0.284
2	0.74	0.80	61.5 / 63.5	0.287 / 0.296
3	0.76	0.77	61.8 / 61.7	0.290 / 0.285
4	0.76	0.78	61.6 / 62.4	0.287 / 0.296
5	0.72	0.75	60.4 / 61.6	0.278 / 0.291
<i>Beta strand, feature 2868 (DSSP strand fraction)</i>				
0	0.001	0.002	49.7 / 59.8	0.152 / 0.251
1	0.029	0.018	49.4 / 52.6	0.154 / 0.186
2	0.039	0.028	50.0 / 55.8	0.157 / 0.215
3	0.033	0.045	50.3 / 58.1	0.163 / 0.235
4	0.065	0.044	50.3 / 59.7	0.161 / 0.249
5	0.110	0.060	50.9 / 57.5	0.167 / 0.226
<i>Coiled coil, feature 813 (pepcoil hit rate)</i>				
0	0%	0%	49.7 / 59.8	0.153 / 0.252
1	0%	0%	59.3 / 65.8	0.253 / 0.303
2	0%	4%	61.7 / 69.5	0.279 / 0.345
3	4%	15%	63.8 / 72.2	0.298 / 0.371
4	3%	19%	65.8 / 73.4	0.307 / 0.386
5	6%	33%	68.1 / 73.0	0.329 / 0.378
<i>Zinc finger, feature 1435 (fraction with any PF00096 zf-C2H2 HMM alignment)</i>				
0	65%	67%	49.7 / 59.8	0.153 / 0.251
1	82%	91%	49.3 / 54.8	0.148 / 0.206
2	81%	94%	51.4 / 56.9	0.165 / 0.224
3	83%	99%	50.4 / 51.7	0.155 / 0.169
4	88%	99%	51.1 / 48.4	0.161 / 0.136
5	82%	93%	51.6 / 47.5	0.167 / 0.123
6	83%	90%	53.9 / 48.1	0.186 / 0.129
7	77%	85%	57.3 / 47.9	0.218 / 0.128
8	91%	93%	59.7 / 48.1	0.237 / 0.130

mean CV (*validation*) AUC, and report its test AUC on shards 4–7. Negatives come from the same proteins as the positives, so a unit cannot win on protein identity. For GLP the candidate bank spans $u \in \{0.1, 0.5, 0.9\}$; all four representations get identical k , subsampling and splits. We report ROC-AUC (threshold-free, the metric of GLP’s released probe); InterPLM’s domain-adjusted F1 would need a per-concept threshold sweep and would refreeze the concept set, so we leave it to future work.

Falsification. For the 94/104 concepts where a meta-neuron beats every single SAE feature, we (i) regress it on the full 10,240-feature SAE dictionary (held-out R^2 : is it a linear read-out?) and (ii) find k^* , the smallest $k \in \{2, 5, 10, 20\}$ at which a joint probe on the SAE’s top- k features reaches its test AUC. Numbers in Table 13.

ESM-2-650M. The scaling point repeats this on the larger host ($d=1280$) at layer 18, against the released **Elana/InterPLM-esm2-650m** SAE, with the GLP trained on 100M activations. The selection procedure is unchanged and retains 166 concepts, a different pool from the 104 at 8M, so only the within-host ordering is comparable (Fig. 13).

D.2 Continuous properties: definition and probe

Concepts. The ten continuous targets are whole-sequence physicochemical scalars computed directly from the amino-acid string, with no annotation lookup and no learned predictor. From Biopython’s **ProteinAnalysis**: GRAVY (mean Kyte–Doolittle hydrophathy), molecular weight, isoelectric point, aromaticity (Phe+Trp+Tyr fraction), instability index (Guruprasad dipeptide-weighted sum), net charge at pH 7, and the helix / turn / sheet secondary-structure fractions (fraction of residues in each structure-prone amino-acid set). From modlAMP’s **GlobalDescriptor**: the aliphatic index (relative side-chain volume of Ala / Val / Ile / Leu). Sequences containing a non-canonical residue (X / B / Z / U / O) are undefined for these scales and dropped.

Probe. Same shards and the same select-on-validation, report-on-test design as the discrete probe, with three changes forced by the target being a per-protein real number rather than a per-residue label: features are per-protein *mean-pooled* activations, not per-residue ones; candidates are ranked by $|\text{Pearson } r|$ with the property; and each 1-D probe is an ordinary least-squares fit scored by Spearman ρ (5-fold out-of-fold on shards 0–3 for selection, shards 4–7 for the reported number). We report test Spearman ρ , not AUC, because a continuous property has no positive / negative split. GLP again pools its candidate bank across $u \in \{0.1, 0.5, 0.9\}$.

D.3 Per-concept tables

Full per-concept/property numbers behind Table 1 and Fig. 14 (Section 3.2).

Table 11: Per-concept test AUC, all 104 discrete Swiss-Prot concepts (24-block/500M model). Bold = best of the four representations.

Concept	GLP	SAE	Raw-layer	Raw-MLP
Domain / motif (n=86)				
Compositional bias_Acidic residues	0.973	0.966	0.922	0.910
Compositional bias_Basic and acidic residues	0.827	0.716	0.786	0.759
Compositional bias_Basic residues	0.946	0.926	0.794	0.891
Compositional bias_Gly residues	0.935	0.874	0.899	0.902
Compositional bias_Low complexity	0.866	0.808	0.849	0.861
Compositional bias_Polar residues	0.869	0.820	0.795	0.876
Compositional bias_Pro residues	0.948	0.889	0.907	0.908
Domain_C-type lectin	0.954	0.891	0.834	0.932
Domain_C1q	0.962	0.843	0.829	0.941
Domain_CTCK	0.757	0.657	0.704	0.751

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Concept	GLP	SAE	Raw-layer	Raw-MLP
Domain_Collagen-like	0.987	0.965	0.978	0.980
Domain_DHHC	0.871	0.855	0.813	0.780
Domain_EF-hand 1	0.662	0.601	0.582	0.615
Domain_EF-hand 2	0.748	0.747	0.677	0.745
Domain_EF-hand 3	0.734	0.714	0.666	0.712
Domain_EF-hand 4	0.675	0.649	0.645	0.696
Domain_EGF-like	0.948	0.925	0.919	0.913
Domain_F-box	0.995	0.966	0.894	0.916
Domain_Fe2OG dioxygenase	0.950	0.806	0.879	0.871
Domain_G-alpha	0.939	0.850	0.782	0.881
Domain_GST C-terminal	0.903	0.737	0.757	0.812
Domain_GST N-terminal	0.980	0.942	0.873	0.953
Domain_Globin	0.912	0.772	0.700	0.864
Domain_H15	0.990	0.932	0.962	0.959
Domain_HTH myb-type 1	0.963	0.911	0.939	0.942
Domain_HTH myb-type 2	0.938	0.852	0.879	0.915
Domain_IGFBP N-terminal	0.759	0.752	0.670	0.758
Domain_Ig-like	0.917	0.868	0.847	0.935
Domain_Ig-like C1-type	0.952	0.884	0.788	0.942
Domain_Ig-like C2-type	0.812	0.704	0.658	0.758
Domain_Ig-like C2-type 1	0.752	0.740	0.712	0.736
Domain_Ig-like C2-type 2	0.772	0.721	0.737	0.794
Domain_Ig-like V-type	0.904	0.875	0.808	0.876
Domain_J	0.971	0.964	0.873	0.950
Domain_K-box	0.941	0.839	0.783	0.893
Domain_LIM zinc-binding 1	0.829	0.791	0.777	0.773
Domain_LIM zinc-binding 2	0.775	0.679	0.718	0.771
Domain_MADS-box	0.909	0.871	0.754	0.919
Domain_MARVEL	0.975	0.817	0.910	0.967
Domain_MPN	0.763	0.763	0.636	0.748
Domain_N-acetyltransferase	0.979	0.920	0.849	0.925
Domain_Nudix hydrolase	0.970	0.882	0.860	0.945
Domain_PB1	0.902	0.676	0.908	0.980
Domain_PDZ	0.967	0.883	0.855	0.974
Domain_PH	0.885	0.799	0.840	0.867
Domain_PPIase cyclophilin-type	0.908	0.793	0.907	0.923
Domain_Peptidase S1	0.981	0.940	0.917	0.986
Domain_Protein kinase	0.964	0.941	0.962	0.968
Domain_RRM	0.994	0.963	0.974	0.987
Domain_RRM 1	0.856	0.826	0.786	0.853
Domain_RRM 2	0.833	0.818	0.820	0.822
Domain_SH2	0.845	0.687	0.751	0.748
Domain_SH3	0.996	0.974	0.914	0.982
Domain_Sm	0.888	0.742	0.842	0.926
Domain_THD	0.916	0.801	0.778	0.893
Domain_TSP type-1	0.677	0.681	0.657	0.781
Domain_Thioredoxin	0.959	0.861	0.867	0.941
Domain_Tyrosine-protein phosphatase	0.880	0.819	0.835	0.816
Domain_UBC core	0.866	0.625	0.680	0.816
Domain_UPAR/Ly6	0.857	0.910	0.810	0.915
Domain_Ubiquitin-like	0.994	0.892	0.879	0.970
Domain_bHLH	0.914	0.811	0.866	0.918
Domain_bZIP	0.982	0.934	0.961	0.981
Domain_sHSP	0.992	0.986	0.976	0.992

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Concept	GLP	SAE	Raw-layer	Raw-MLP
Domain_t-SNARE coiled-coil homology	0.886	0.843	0.786	0.841
Domain_v-SNARE coiled-coil homology	0.971	0.964	0.893	0.957
Motif_Nuclear localization signal	0.826	0.881	0.741	0.837
Motif_Nudix box	0.996	0.978	0.956	0.991
Motif_Switch 1	0.990	0.971	0.938	0.984
Motif_Switch 2	0.980	0.946	0.931	0.969
Region_Alpha-1	0.865	0.740	0.753	0.848
Region_Alpha-2	0.811	0.700	0.847	0.810
Region_Alpha-3	0.994	0.959	0.830	0.985
Region_Basic motif	0.983	0.986	0.954	0.988
Region_Disordered	0.884	0.776	0.885	0.878
Region_Helix-loop-helix motif	0.942	0.891	0.936	0.930
Region_LID	0.886	0.799	0.839	0.827
Region_Leucine-zipper	0.936	0.905	0.883	0.933
Region_NMP	0.937	0.935	0.911	0.959
Region_Switch-II	0.973	0.963	0.942	0.934
Zinc finger_C2H2-type 1	0.850	0.806	0.804	0.871
Zinc finger_C2H2-type 2	0.821	0.790	0.773	0.788
Zinc finger_C2H2-type 3	0.844	0.743	0.787	0.834
Zinc finger_C4-type	0.713	0.721	0.619	0.567
Zinc finger_RING-type	0.966	0.948	0.888	0.961
Zinc finger_any	0.927	0.873	0.835	0.923
Chemical / PTM (n=9)				
Disulfide bond	0.997	0.978	0.921	0.974
Glycosylation_N-linked (GlcNAc...)	0.993	0.982	0.906	0.982
asparagine				
Glycosylation_any	0.960	0.925	0.881	0.935
Lipidation	0.868	0.810	0.912	0.886
Modified residue_N6-acetyllysine	0.963	0.967	0.914	0.958
Modified residue_N6-succinyllysine	0.953	0.956	0.888	0.937
Modified residue_Phosphoserine	0.975	0.974	0.867	0.944
Modified residue_Phosphothreonine	0.980	0.987	0.896	0.839
Modified residue_Phosphotyrosine	0.976	0.979	0.922	0.981
Structural (n=4)				
Beta strand	0.768	0.629	0.616	0.668
Coiled coil	0.878	0.837	0.759	0.826
Helix	0.780	0.618	0.629	0.700
Turn	0.604	0.556	0.553	0.594
Topology / localization (n=3)				
Transit peptide_Chloroplast	0.978	0.975	0.875	0.959
Transit peptide_Mitochondrion	0.921	0.895	0.899	0.873
Transit peptide_any	0.944	0.929	0.904	0.907
Binding / catalytic (n=2)				
Active site_Proton acceptor	0.773	0.691	0.775	0.846
Active site_any	0.715	0.674	0.725	0.688

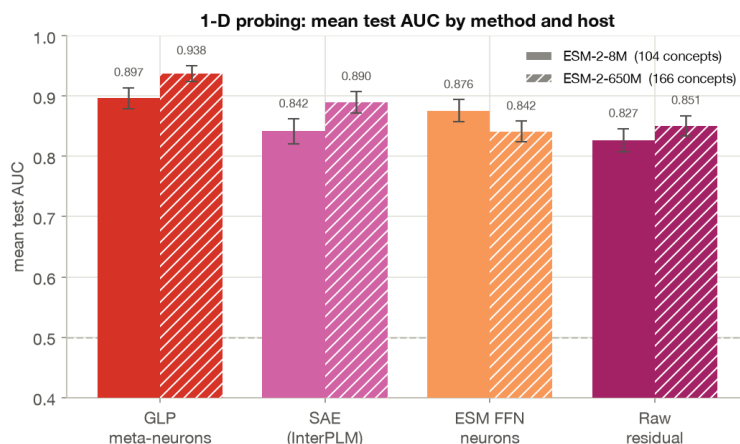


Figure 13: 1-D probing mean test AUC by method and host (ESM-2-8M solid, ESM-2-650M hatched; 95% bootstrap CI). GLP meta-neurons lead at both host sizes. The two hosts use different concept pools (104 vs 166), so absolute AUCs are not comparable across hosts; the within-host ordering is. ESM’s FFN neurons drop from second (8M) to last (650M) as the SAE strengthens.

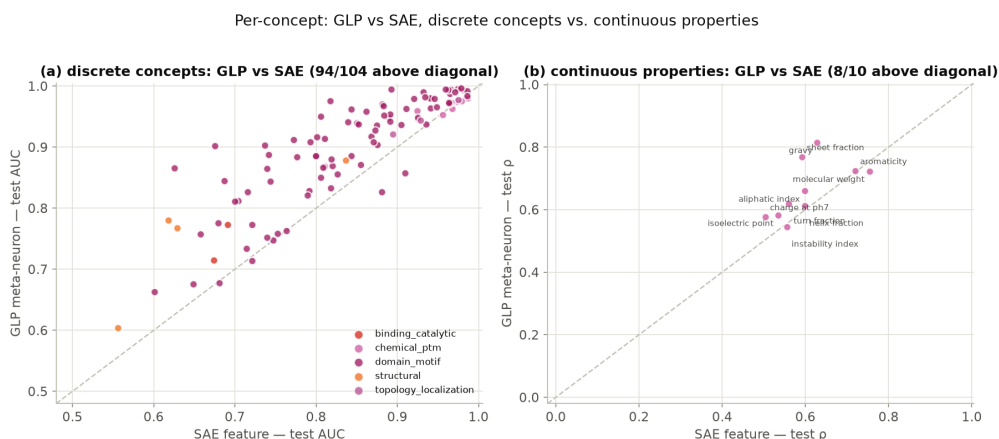


Figure 14: Per-concept/property GLP vs. SAE. Left: discrete concepts, colored by Swiss-Prot category, most points sit above the diagonal (GLP wins), across categories, not concentrated in one bucket (the pool is dominated by domain/motif, 86/104, so a category-balanced re-run remains future work, Section ??). Right: continuous properties, direct-labeled, GLP wins 8/10, led by gravity and sheet fraction.

Table 13: Falsification check, all 94 concepts where a GLP meta-neuron beats every SAE feature: the winning meta-neuron, the held-out R^2 of regressing its activation on the SAE’s full 10,240-feature dictionary, and k^* , the smallest $k \in \{2, 5, 10, 20\}$ at which a joint logistic probe over the SAE’s own top- k features reaches GLP’s test AUC (“never” = no $k \leq 20$ does). Sorted by k^* (“never” first, the 15-concept discovery list), then by R^2 ascending within each k^* group, deliberately not by R^2 alone, since Section 3.2 shows R^2 does not order this list the way it might suggest. R^2 distribution: mean 0.752, median 0.756, IQR [0.690, 0.841], 36% above 0.8.

Concept	Winning meta-neuron (u , unit)	R^2	k^*
Domain_Ubiquitin-like	$u=0.9$, unit 14140	0.522	never

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Concept

Winning meta-
neuron (u , unit) R^2 k^*

Domain_sHSP	$u=0.9$, unit 19309	0.688	never
Domain_F-box	$u=0.9$, unit 15083	0.690	never
Motif_Nudix box	$u=0.9$, unit 21801	0.695	never
Active site_Proton acceptor	$u=0.5$, unit 26160	0.704	never
Domain_EF-hand 3	$u=0.9$, unit 20526	0.729	never
Domain_Thioredoxin	$u=0.9$, unit 29393	0.743	never
Domain_Nudix hydrolase	$u=0.9$, unit 16904	0.751	never
Beta strand	$u=0.9$, unit 16205	0.757	never
Domain_MARVEL	$u=0.5$, unit 27041	0.788	never
Domain_RRM 1	$u=0.9$, unit 7843	0.833	never
Domain_SH3	$u=0.9$, unit 20077	0.843	never
Domain_GST C-terminal	$u=0.9$, unit 30479	0.858	never
Domain_N-acetyltransferase	$u=0.9$, unit 19170	0.896	never
Disulfide bond	$u=0.9$, unit 10335	0.963	never
Compositional bias_Acidic residues	$u=0.9$, unit 8455	0.514	2
Domain_EF-hand 4	$u=0.9$, unit 12173	0.596	2
Domain_IGFBP N-terminal	$u=0.9$, unit 22611	0.703	2
Domain_J	$u=0.9$, unit 27480	0.730	2
Domain_DHHC	$u=0.9$, unit 17455	0.765	2
Transit peptide_Mitochondrion	$u=0.9$, unit 13498	0.819	2
Region_NMP	$u=0.9$, unit 26894	0.831	2
Region_Leucine-zipper	$u=0.9$, unit 6707	0.867	2
Transit peptide_any	$u=0.9$, unit 13480	0.872	2
Compositional bias_Basic residues	$u=0.9$, unit 1776	0.931	2
Active site_any	$u=0.9$, unit 4948	0.953	2
Modified residue_Phosphoserine	$u=0.9$, unit 4206	0.960	2
Domain_PDZ	$u=0.9$, unit 23229	0.497	5
Domain_THD	$u=0.9$, unit 4078	0.563	5
Domain_PB1	$u=0.9$, unit 13975	0.575	5
Domain_t-SNARE coiled-coil homology	$u=0.9$, unit 16644	0.612	5
Region_Switch-II	$u=0.9$, unit 25714	0.687	5
Domain_bHLH	$u=0.9$, unit 19767	0.718	5
Domain_EF-hand 1	$u=0.9$, unit 23994	0.720	5
Coiled coil	$u=0.9$, unit 9432	0.750	5
Compositional bias_Polar residues	$u=0.9$, unit 590	0.756	5
Domain_bZIP	$u=0.9$, unit 20128	0.758	5
Compositional bias_Pro residues	$u=0.9$, unit 21640	0.762	5
Zinc finger_C2H2-type 2	$u=0.9$, unit 25886	0.773	5
Domain_Ig-like C2-type 1	$u=0.9$, unit 26790	0.777	5
Domain_Tyrosine-protein phosphatase	$u=0.9$, unit 14390	0.780	5
Domain_v-SNARE coiled-coil homology	$u=0.9$, unit 21460	0.790	5
Compositional bias_Low complexity	$u=0.9$, unit 30664	0.805	5
Domain_Collagen-like	$u=0.9$, unit 10974	0.806	5
Region_Alpha-1	$u=0.9$, unit 9498	0.837	5
Zinc finger_any	$u=0.9$, unit 22205	0.850	5
Zinc finger_C2H2-type 1	$u=0.9$, unit 27348	0.855	5
Domain_K-box	$u=0.9$, unit 6707	0.867	5
Domain_Peptidase S1	$u=0.9$, unit 28973	0.870	5
Domain_Ig-like V-type	$u=0.9$, unit 29950	0.879	5
Domain_Ig-like	$u=0.9$, unit 29950	0.879	5
Domain_Protein kinase	$u=0.9$, unit 4491	0.906	5
Lipidation	$u=0.9$, unit 6334	0.918	5
Domain_MADS-box	$u=0.9$, unit 9776	0.945	5
Glycosylation_N-linked (GlcNAc...) asparagine	$u=0.5$, unit 4948	0.956	5
Domain_Sm	$u=0.5$, unit 20914	0.332	10
Helix	$u=0.9$, unit 3472	0.546	10
Domain_Globin	$u=0.9$, unit 13975	0.575	10
Domain_PPIase cyclophilin-type	$u=0.9$, unit 18688	0.588	10
Region_LID	$u=0.9$, unit 26724	0.601	10
Motif_Switch 2	$u=0.5$, unit 26521	0.619	10

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Concept	Winning neuron (u , unit)	R^2	k^*
Zinc finger_RING-type	$u=0.5$, unit 24926	0.622	10
Domain_EGF-like	$u=0.9$, unit 15026	0.644	10
Domain_G-alpha	$u=0.9$, unit 18912	0.655	10
Domain_CTCK	$u=0.9$, unit 12361	0.687	10
Compositional bias_Basic and acidic residues	$u=0.9$, unit 8044	0.690	10
Turn	$u=0.9$, unit 6039	0.695	10
Domain_EF-hand 2	$u=0.9$, unit 7980	0.702	10
Domain_RRM	$u=0.9$, unit 24887	0.731	10
Domain_GST N-terminal	$u=0.5$, unit 30322	0.734	10
Domain_Ig-like C2-type	$u=0.9$, unit 12319	0.737	10
Domain_SH2	$u=0.9$, unit 12698	0.739	10
Domain_Fe2OG dioxygenase	$u=0.9$, unit 24008	0.786	10
Region_Alpha-3	$u=0.9$, unit 9122	0.801	10
Region_Disordered	$u=0.9$, unit 29328	0.831	10
Domain_H15	$u=0.9$, unit 23219	0.851	10
Compositional bias_Gly residues	$u=0.9$, unit 633	0.875	10
Glycosylation_any	$u=0.9$, unit 5872	0.949	10
Domain_LIM zinc-binding 2	$u=0.9$, unit 23043	0.411	20
Domain_LIM zinc-binding 1	$u=0.9$, unit 19181	0.670	20
Domain_C-type lectin	$u=0.9$, unit 12361	0.687	20
Region_Alpha-2	$u=0.9$, unit 12361	0.687	20
Domain_C1q	$u=0.9$, unit 10286	0.706	20
Motif_Switch 1	$u=0.9$, unit 26867	0.730	20
Domain_PH	$u=0.9$, unit 28698	0.733	20
Domain_UBC core	$u=0.9$, unit 12698	0.739	20
Region_Helix-loop-helix motif	$u=0.9$, unit 21956	0.751	20
Domain_HTH myb-type 2	$u=0.9$, unit 20184	0.762	20
Zinc finger_C2H2-type 3	$u=0.9$, unit 28162	0.765	20
Domain_HTH myb-type 1	$u=0.9$, unit 14786	0.798	20
Domain_RRM 2	$u=0.9$, unit 25236	0.804	20
Domain_Ig-like C2-type 2	$u=0.9$, unit 8105	0.818	20
Transit peptide_Chloroplast	$u=0.9$, unit 13480	0.872	20
Domain_Ig-like C1-type	$u=0.9$, unit 23547	0.892	20

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Table 12: Per-property test Spearman ρ , all 10 continuous sequence properties (24-block/500M model). Bold = best of the four representations.

Property	GLP	SAE	Raw-layer	Raw-MLP
aliphatic index	0.660	0.598	0.621	0.563
aromaticity	0.724	0.720	0.479	0.507
charge at ph7	0.577	0.504	0.362	0.595
gravy	0.815	0.628	0.519	0.646
helix fraction	0.612	0.598	0.290	0.499
instability index	0.546	0.556	0.465	0.545
isoelectric point	0.582	0.535	0.380	0.599
molecular weight	0.723	0.754	0.447	0.461
sheet fraction	0.769	0.592	0.632	0.678
turn fraction	0.620	0.560	0.491	0.535